

The Cost of Late Delivery: A Data Driven Analysis of Olist Customer Reviews

September 2016 – October 2018

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Tools: Python | PostgreSQL | Microsoft Word

Executive Summary

This report analyzes 99,441 Olist orders between September 2016 and October 2018 to assess how delivery performance influences customer satisfaction, measured through customer review scores.

The analysis found a strong relationship between delivery timing and customer ratings. Orders delivered late received an average review score of 2.27, compared with 4.29 for orders delivered early—a difference of just over two points on a five-point scale. Statistical testing confirmed that this relationship was highly significant ($p < 0.001$), and the same pattern was observed across product categories.

These findings indicate that delivery reliability is a platform-wide operational issue rather than one limited to specific products. Reducing late deliveries, identifying the operational causes of delays, and improving delivery estimates and customer communication represent practical opportunities to improve customer experience across the marketplace.

Background & Objectives

Olist is a Brazilian e-commerce marketplace that enables small and medium-sized businesses to sell through major online channels while managing logistics and customer relationships on their behalf. As a marketplace business, customer satisfaction is a critical performance indicator because it influences seller reputation, customer retention, and the long-term health of the platform.

Delivery performance is one of the few parts of the customer journey that every buyer experiences, making it a key operational driver of customer perception. Delays can undermine the purchasing experience regardless of product quality, while reliable delivery has the potential to strengthen trust and improve overall satisfaction. Understanding the extent of this relationship is therefore important for operational and strategic decision-making.

This analysis evaluates whether the timing of an order's delivery—early, on time, or late relative to the estimated delivery date—has a measurable impact on customer review scores. Specifically, it addresses two questions:

- ✓ Does delivery timing have a statistically significant relationship with customer review scores?
- ✓ Is this relationship consistent across product categories, or does it vary by product type?

The findings provide an evidence-based assessment of whether improving delivery reliability is likely to increase customer satisfaction across the marketplace and whether operational improvement efforts should be targeted at specific categories or applied more broadly.

Methodology

The analysis was conducted using nine Olist datasets covering orders, customers, order items, payments, products, reviews, sellers, and geolocation data. Each dataset underwent a structured data quality assessment to identify and address issues such as missing values, duplicate records, inconsistent order statuses, and invalid geolocation records before being integrated into the analysis.

Following data preparation, the datasets were consolidated into a single order-level dataset, combining customer, payment, delivery, product, and review information for each order. The analysis was limited to completed orders with valid customer reviews, ensuring that both delivery performance and customer satisfaction could be measured consistently.

Delivery performance was evaluated by calculating the difference, in days, between the actual delivery date and the estimated delivery date. Orders were then classified as early, on time, or late, allowing customer review scores to be compared across delivery outcomes. The analysis examined these relationships at both the overall marketplace level and within individual product categories to determine whether the effect of delivery timing varied across different types of products.

Statistical testing was used to distinguish meaningful patterns from random variation. A one-way ANOVA assessed differences in review scores across all three delivery groups, an independent samples t-test compared early and late deliveries, and a Pearson correlation measured the relationship between delivery delay (in days) and customer review scores.

The analysis was performed in Python using pandas for data preparation and aggregation and SciPy for statistical testing. The core analysis was also repeated in PostgreSQL using equivalent data cleaning and aggregation logic. The results were consistent with the Python analysis, providing an independent validation of the findings

Key Findings

Table 1 summarises review scores and order volumes across the three delivery outcomes.

Delivery Timing	Average Review Score	Number of Orders	Share of Orders
Early	4.29	88,168	92.0%
On Time	4.03	1,280	1.3%
Late	2.29	6,382	6.7%

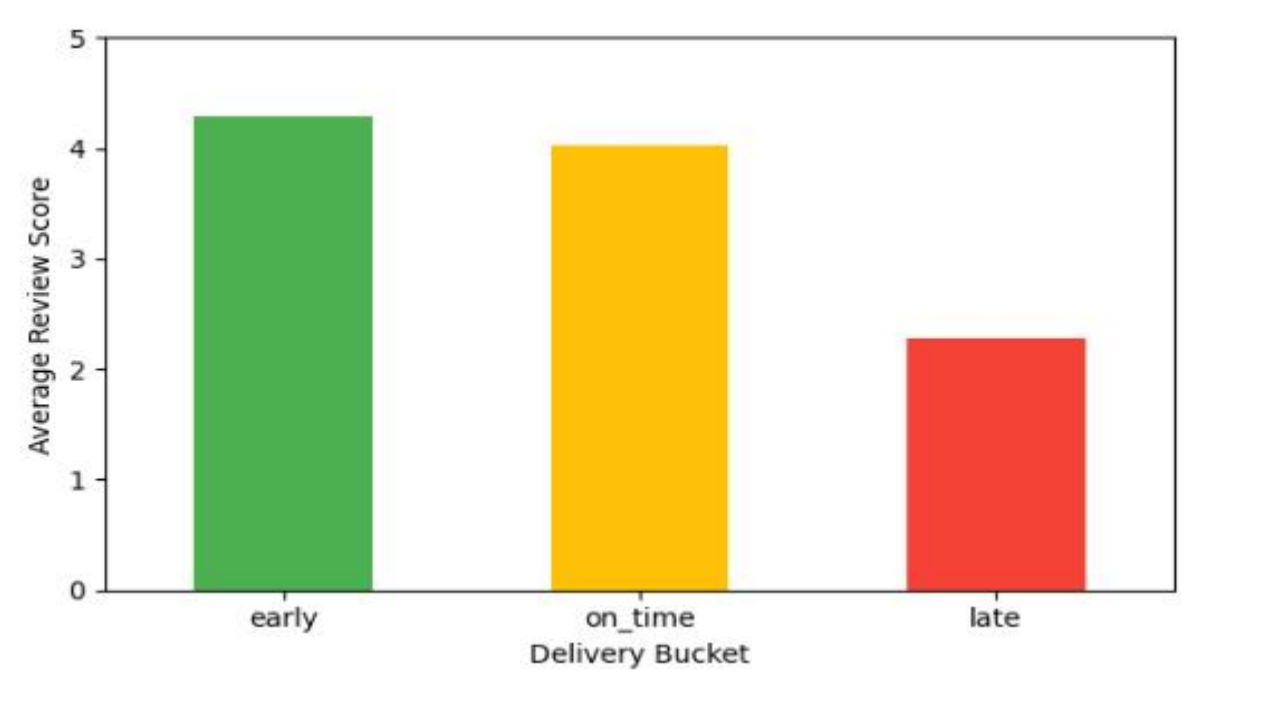


Figure 1: Average Review Score by Delivery Bucket

Orders delivered late received an average review score of 2.27, compared with 4.29 for orders delivered early—a difference of just over two points on a five-point scale. Orders delivered on the estimated date averaged 4.03, indicating that customers rated early deliveries slightly more positively than deliveries that simply met expectations.

More than 92% of orders in the dataset were delivered before the estimated delivery date, indicating that Olist's delivery estimates are generally conservative. Although late deliveries accounted for a relatively small share of

total orders, they were consistently linked to much lower customer ratings, making delivery delays a high-impact driver of customer dissatisfaction.

Statistical Significance

The relationship between delivery timing and customer review scores was tested using three independent statistical methods, all of which produced consistent results:

- ✓ A one-way ANOVA comparing early, on-time, and late deliveries found a statistically significant difference in average review scores across the three groups ($F = 8,724.86$, $p < 0.001$).
- ✓ An independent samples t-test comparing early and late deliveries reached the same conclusion ($t = 100.91$, $p < 0.001$).
- ✓ A Pearson correlation analysis also identified a moderate negative relationship between delivery delay (measured in days) and review score ($r = -0.267$, $p < 0.001$), indicating that longer delivery delays were generally associated with lower customer ratings. The moderate correlation suggests that delivery timing is an important influence on customer satisfaction, but not the only factor affecting review scores.

Consistency across product categories

To assess whether this relationship differed by product type, the analysis was repeated across 22 product categories with sufficient order volume. In every category, customers rated late deliveries lower than early deliveries. The difference ranged from approximately 1.4 to 2.3 review points, demonstrating that the relationship was consistent across the marketplace rather than being driven by a small number of categories:

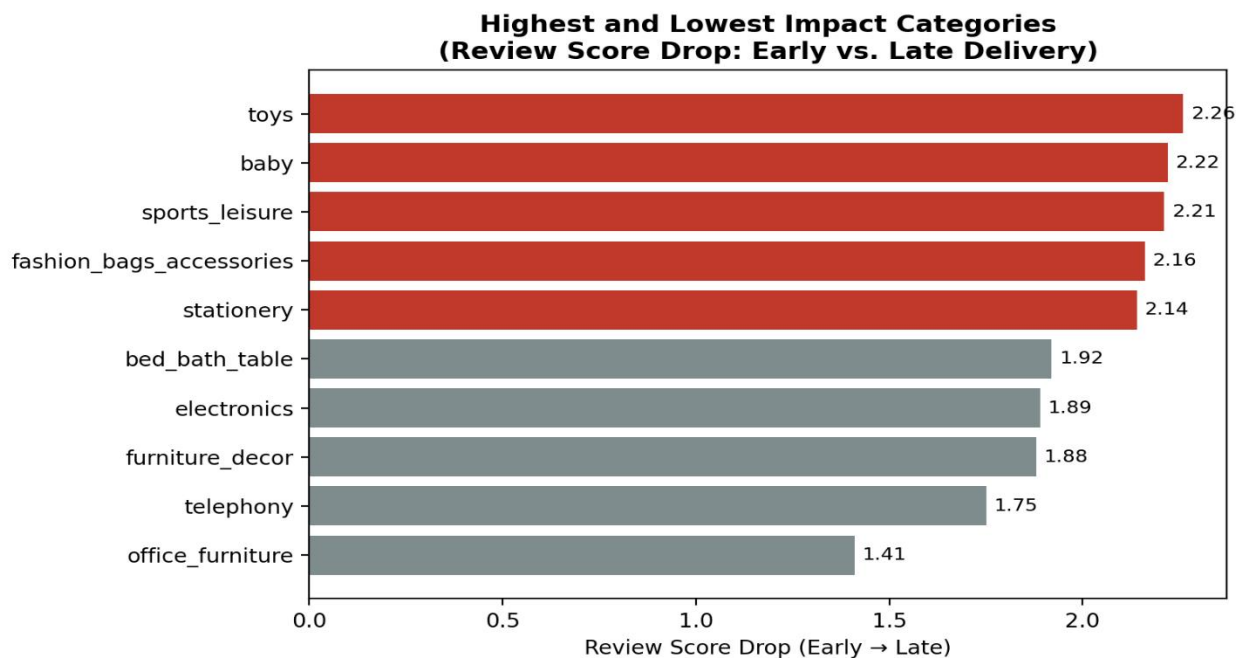


Figure 2: Highest and Lowest Impact Categories — Review Score Drop, Early vs. Late Delivery

The largest declines were observed in toys, baby products, and sports & leisure, where average review scores fell by more than 2.2 points between early and late deliveries. Office furniture and telephony recorded smaller declines, ranging from approximately 1.4 to 1.75 points. Office furniture also had the lowest average review score for early deliveries (3.76), compared with roughly 4.15 to 4.45 across most other categories. This suggests that

factors beyond delivery performance may be affecting customer experience in that category and merit separate investigation.

Taken together, these findings indicate that the effect of delivery timing is consistent across product categories. While the magnitude of the impact varies slightly, late delivery is associated with lower customer ratings throughout the marketplace rather than within only a small subset of products.

Recommendations

The analysis indicates that improving delivery reliability offers one of the clearest opportunities to increase customer satisfaction across the Olist marketplace. Based on the findings, the following actions are recommended:

1. Set a measurable target for reducing late deliveries. Late deliveries received an average review score of 2.27, compared with 4.29 for orders delivered early. Although only around 6.7% of delivered and reviewed orders arrived late, they accounted for a small share of orders but a large share of low customer ratings. Reducing the frequency of late deliveries should therefore be established as a key operational performance objective.

2. Identify the operational drivers of late deliveries. Since more than 92% of orders were delivered before the estimated date, the remaining late deliveries are likely to reflect specific operational issues rather than a platform-wide capacity constraint. Analysis at the carrier, seller, fulfilment centre, and regional levels could help identify where delays originate and allow improvement efforts to be targeted more effectively.

3. Prioritize delivery reliability across the marketplace rather than by product category. The relationship between delivery timing and customer ratings was consistent across all 22 product categories analysed. This suggests that broad improvements to delivery performance are likely to generate greater returns than category-specific delivery initiatives.

4. Investigate customer satisfaction within the office furniture category. Office furniture recorded the lowest average review score even when orders were delivered early, indicating that delivery performance alone does not explain customer feedback. Additional analysis should examine factors such as product quality, pricing, packaging, or seller performance to identify the underlying causes.

5. Strengthen communication for orders at risk of delay. While preventing delays should remain the priority, customers may respond more positively when expectations are managed proactively. Providing timely updates and revised delivery estimates for orders likely to arrive late could help reduce the negative impact on customer experience.

6. Review the accuracy of estimated delivery dates. More than 92% of orders arrived before the estimated delivery date, suggesting that current delivery estimates are generally conservative. Refining delivery estimates to better reflect actual performance may improve customer expectations while maintaining a realistic buffer against unforeseen delays.

Limitations

The findings should be interpreted alongside several limitations:

First, the analysis identifies a statistical relationship between delivery timing and customer review scores but does not establish causation. Other factors, including product quality, pricing, seller performance, and customer expectations, may also influence review scores and were not examined as part of this study.

Second, the Pearson correlation between delivery delay and review score ($r = -0.267$) indicates a moderate relationship. Delivery timing is therefore an important factor in customer satisfaction, but it is only one of several influences on how customers rate their experience.

Third, late deliveries accounted for approximately 6.7% of delivered and reviewed orders, making this group substantially smaller than the early delivery group.

Although the sample size remained sufficient for statistical testing, estimates relating to late deliveries are based on fewer observations than those for early deliveries.

Finally, the analysis is based on Olist order data collected between September 2016 and October 2018. Operational processes, logistics networks, and customer behaviour may have changed since then, so the findings should be interpreted within the context of the period analysed.

Conclusion

This analysis examined whether delivery timing influences customer satisfaction on the Olist marketplace. Across 95,830 delivered and reviewed orders, the results consistently showed that customers gave significantly lower review scores to late deliveries than to orders delivered on time or ahead of schedule. Statistical testing confirmed that this relationship was unlikely to have occurred by chance, and the same pattern was observed across all 22 product categories included in the analysis.

Although delivery timing is only one of several factors that influence customer satisfaction, it represents an operational area where improvements can be measured and monitored over time. The consistency of the findings across product categories suggests that reducing late deliveries would benefit the marketplace broadly rather than a limited group of products.

The recommendations presented in this report focus on reducing late deliveries, identifying the operational causes of delays, improving communication with affected customers, and reviewing estimated delivery dates. Together, these actions provide a practical framework for improving delivery performance and strengthening the customer experience across the platform.